6. Suppose we collect data for a group of students in a statistics class with variables $X_1 =$ hours studied, $X_2 =$ undergrad GPA, and $Y =$ receive an A. We fit a logistic regression and produce estimated coefficient, $\hat{\beta}_0 = -6, \hat{\beta}_1 = 0.05, \hat{\beta}_2 = 1$.

(a) Estimate the probability that a student who studies for 40 h and has an undergrad GPA of 3.5 gets an A in the class.

X_1 = hour studied

X_2 = undergrad GPA

$$p = \sigma(\mu) = \frac{1}{1+e^{-\mu}} \qquad \mu = \underset{-6}{\beta_0} + \underset{0.05}{\beta_1} X_1 + \underset{1}{\beta_2} X_2$$

(a) $X_1 = 40$

$X_2 = 3.5$

$\mu = -6 + 0.05 \times 40 + 1 \times 3.5 = -0.5$

$$p = \sigma(0.5) = \frac{1}{1+e^{-0.5}} = \frac{1}{1+1.6487} \approx 0.3775 = 37.8\% \#$$

(b) How many hours would the student in part (a) need to study to have a 50 % chance of getting an A in the class?

$$p = \sigma(\mu) = \frac{1}{1+e^{\mu}} = 0.5 \qquad X_2 = 3.5$$

$1 + e^{\mu} = 2 \qquad e^{\mu} = 1 \qquad \mu = 0$

$0 = -6 + 0.05 \times X_1 + 1 \times 3.5$

$X_1 = 50$ hour #